Mississippi Valley State University

CS 341 — Discrete Structures

Practice Set 18: Approved Solutions

Functions, Range, and Composition

Covers Chartrand & Zhang, Section 5.3

Status of this set

Ungraded practice. Solutions released with it. Attempt everything first.

Part I: How to Practice This Material

Check both failure modes

Is it a function?

A relation $g \subseteq A \times B$ fails to be a function from A to B if

1. some element of A has **no** image, or
2. some element of A has **more than one**.

Check both. Repeated values in B are fine — only the A side is constrained.

Three words, kept straight

Domain = where inputs come from. **Codomain** = the declared target set. **Range** = what actually gets hit.
Range $\subseteq$ codomain always, and equality is a special condition with its own name ("onto," on Wednesday).

Three mistakes to expect

1. **Confusing range with codomain.** The codomain is declared; the range is computed.
2. **Reading $g \circ f$ left to right.** f is applied first.
3. **Assuming $f \circ g = g \circ f$.** They are usually different.

Part II: Practice Problems

Section A — Is it a function?

1. Let $A = \{1, 2, 3, 4\}$ and $B = \{p, q, r\}$. For each relation, decide whether it is a function from A to B, with a reason.

(a) $\{(1, p), (2, q), (3, p), (4, r)\}$

(b) $\{(1, p), (2, q), (3, r)\}$

(c) $\{(1, p), (1, q), (2, r), (3, p), (4, q)\}$

(d) $\{(1, q), (2, q), (3, q), (4, q)\}$

Approved solution

(a) Yes. Each of $1, 2, 3, 4$ appears exactly once as a first coordinate.
(b) No. The element 4 is not a first coordinate of any pair, so it has no image.
(c) No. The element 1 appears twice, with images p and q, so its image is not unique.
(d) Yes. A constant function. Every element of A has exactly one image; the fact that they all share it is irrelevant.

(b) and (c) are the two failure modes — one missing, one duplicated. **(d) is the check on whether students have absorbed that only the domain side is restricted.**

Section B — Domain, codomain, range

2. Let $A = \{1, 2, 3, 4, 5\}$, $B = \{a, b, c, d\}$, and $f = \{(1, b), (2, d), (3, b), (4, a), (5, d)\}$.

(a) State the domain, codomain, and range of f.

(b) Compute $f(\{1, 3\})$ and $f(\{2, 4\})$.

(c) Is there a subset $X \subseteq A$ with $|X| = 2$ and $|f(X)| = 1$? Give one or explain why not.

Approved solution

(a) Domain $= A = \{1, 2, 3, 4, 5\}$. Codomain $= B = \{a, b, c, d\}$. Range $= f(A) = \{a, b, d\}$ — note c is the image of nothing.
(b) $f(\{1, 3\}) = \{b\}$, since $f(1) = f(3) = b$. $f(\{2, 4\}) = \{d, a\} = \{a, d\}$.
(c) Yes: $X = \{1, 3\}$ gives $f(X) = \{b\}$, so $|X| = 2$ and $|f(X)| = 1$. Also $X = \{2, 5\}$ works.

Part (b) shows that $|f(X)|$ can be smaller than $|X|$ but never larger. Part (c) is exactly the situation that makes f fail to be one-to-one — Wednesday's topic, arriving early.

Section C — Composition

3. Let $A = \{1, 2, 3\}$, $B = \{w, x, y\}$, $C = \{5, 6, 7\}$, with

$$f = \{(1, x), (2, w), (3, x)\}, \qquad g = \{(w, 6), (x, 5), (y, 7)\}.$$

Compute $g \circ f$ as a set of ordered pairs, and give its range.

Approved solution

$(g \circ f)(1) = g(f(1)) = g(x) = 5$. $(g \circ f)(2) = g(w) = 6$. $(g \circ f)(3) = g(x) = 5$.
So $g \circ f = \{(1,5),(2,6),(3,5)\}$, with range $\{5,6\}$.
Note 7 is in the codomain but not the range: nothing maps to y under f, so g never gets the chance to send anything to 7.
Note also that $f \circ g$ is not defined here, since the codomain of g is C and the domain of f is A — they do not match up. Composition requires the pieces to fit.

4. Let $f, g : \mathbb{R} \to \mathbb{R}$ with $f(x) = x - 4$ and $g(x) = x^2$.

(a) Find formulas for $g \circ f$ and $f \circ g$.

(b) Evaluate both at $x = 1$ and at $x = 4$.

(c) Are there any values of x where the two agree? Find them.

Approved solution

(a) $(g \circ f)(x) = g(x-4) = (x-4)^2$. $(f \circ g)(x) = f(x^2) = x^2 - 4$.
(b) At $x = 1$: $(g \circ f)(1) = 9$ and $(f \circ g)(1) = -3$.
At $x = 4$: $(g \circ f)(4) = 0$ and $(f \circ g)(4) = 12$.
(c) Set $(x-4)^2 = x^2 - 4$. Expanding, $x^2 - 8x + 16 = x^2 - 4$, so $-8x = -20$ and $x = \frac{5}{2}$.
Check: $(2.5 - 4)^2 = 2.25$ and $2.5^2 - 4 = 2.25$. ✓

Part (c) is the useful one. The two compositions agree at exactly one point, which is a sharper statement than "they are different." Two functions that agree somewhere are still different functions — agreement at a point is not equality.

Section D — Writing

5. In five to eight complete sentences, explain the difference between the codomain and the range of a function, and why the codomain must be stated rather than computed. Give a concrete example where two functions share a rule and a domain but differ in codomain.

Approved solution

Model answer. The domain and codomain are both part of the declaration of a function: writing $f : A \to B$ announces that inputs come from A and that outputs are promised to lie in B. The range, by contrast, is computed — it is the set of values actually taken, $f(A) = \{f(a) : a \in A\}$, and it is always a subset of the codomain but need not equal it. The codomain cannot be recovered from the rule, because many different target sets contain all the outputs; declaring it is a choice the definition makes. For example, $f : \mathbb{R} \to \mathbb{R}$ given by $f(x) = x^2$ and $g : \mathbb{R} \to [0, \infty)$ given by the same rule have the same domain and produce the same outputs, but they are different functions because their codomains differ. That difference matters: g is onto and f is not, even though neither their rules nor their ranges differ at all.

Full credit requires the point that the codomain is *declared*. An answer saying only "the range is a subset of the codomain" earns about half — it states the relationship without explaining why the codomain is not determined by the rule.

Section E — Challenge (optional)

6. Let A and B be finite sets with $|A| = m$ and $|B| = n$. How many functions are there from A to B? Justify.

Approved solution

n^m.

Each of the m elements of A must be assigned exactly one image, and the choices are independent: element one has n options, element two has n options, and so on through all m elements. Multiplying gives $n \cdot n \cdots n = n^m$.

Check against Lecture 18. With $|A| = 2$ and $|B| = 3$ this predicts $3^2 = 9$ functions — and Chartrand's Exercise 9 lists exactly nine.

Compare with two earlier counts. There are 2^{mn} *relations* from A to B, since a relation is any subset of $A \times B$. Only n^m of those are functions. For $m = n = 3$ that is 512 relations and 27 functions. Once again the answer is a product of independent choices — the same argument as the Day 1 password count, now for the fourth time.

Instructor notes

- **Problem 1(d)** — the constant function — checks whether students think repeated outputs disqualify a relation. Some will say no.
- **Problem 2(c)** quietly previews one-to-one. Worth pointing at on Wednesday.
- **Problem 4(c)** is better than the usual "show they differ": solving for where they agree gives a single point and makes the distinction between *equal functions* and *equal at a point* concrete.
- **Problem 5** is the writing anchor. The claim that the codomain is declared rather than computed is what makes Wednesday's onto material coherent — if it does not land here, Lecture 19 Section 4 will not either.
- Reserved for Homework 8: Section 5.3 Exercises 2, 4, 14, 30 and Section 5.4 even exercises. Odd ones have printed answers and are self-check only.